

MILK TRANSPORTATION

NSD produces about 20,000 gallons of milk per day, which is transported to Davisco's Le Sueur Cheese plant each day in just over 3 milk tanker trailers. NSD's milk results in the production of approximately 22,000 lbs. of cheese each day. In addition, whey proteins and lactose are harvested at the cheese plant and are supplied to many companies that utilize these ingredients in food, nutraceutical and pharmaceutical products that are consumed in the United States and around the world. All equipment used at NSD that comes into contact with milk is made of food-grade stainless steel and/or other materials approved for use as food contact surfaces.



UNIVERSITY OF MINNESOTA

New Sweden Dairy has an affiliation with the University of Minnesota College of Veterinary Medicine. The University utilizes the dairy facilities for housing, teaching and demonstration in their program of teaching veterinary students, continuing education of industry professionals, and researching emerging products and/or practices. The relationship is a public-private partnership that exploits the strengths of the academic knowledge the University teaches and researches as well as the commercial scope, scale and practices that the dairy business (NSD) must employ to be successful. The interaction facilitates an environment that merges the skill sets of the University and of New Sweden Dairy into the development of the necessary knowledge to encourage the continual improvement important to all aspects of milk production, quality, and food safety from the farm to the fork.

Thank you for taking the time to learn about New Sweden Dairy. We encourage you and your family to support the production of milk by consuming the many wholesome dairy products that are made from and with milk.

New Sweden Dairy Mission Statement

To build and operate a state-of-the-art dairy facility which is operated in a manner that meets or exceeds existing mandated regulatory requirements, optimizes animal health and comfort, provides opportunity for fulfilling employment, and produces the highest quality dairy products possible.



Dairy Education Center College of Veterinary Medicine University of Minnesota Mission Statement

1. To educate veterinary students in the care and management of dairy cows.
2. To educate practicing veterinarians and other professionals serving the dairy industry.
3. To serve as a site for applied research and development.
4. To demonstrate excellent animal care, environmental protection, sustainability and efficient and high quality food production.

New Sweden Dairy (LLP)

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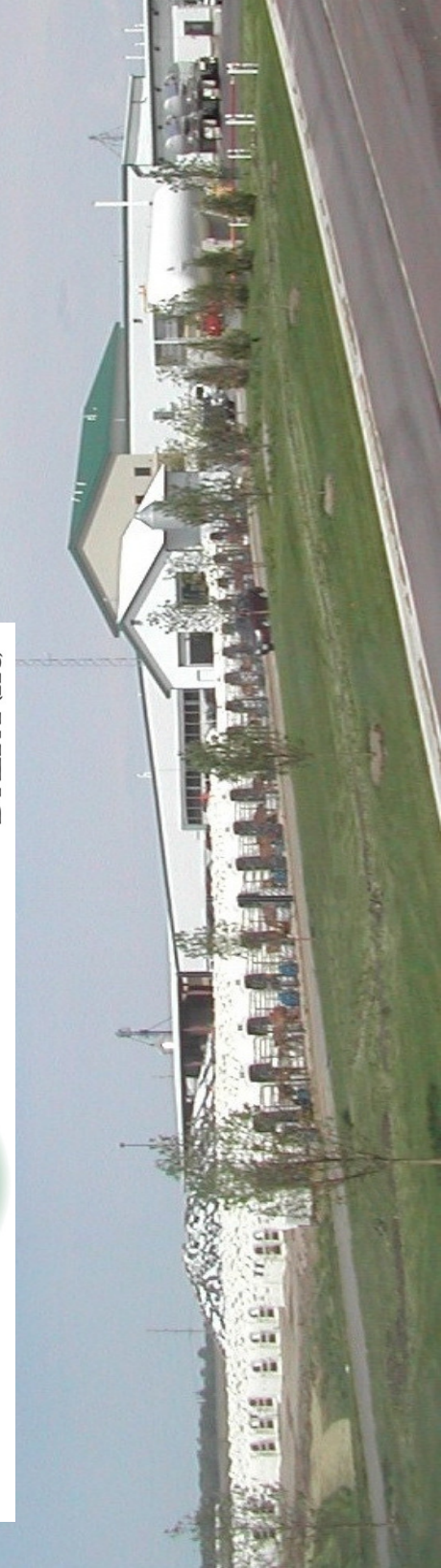
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NEW SWEDEN DAIRY

New Sweden Dairy (NSD) is owned and operated by the Davis family from St. Peter, MN. The family also owns and operates Northern Plains Dairy, Davisco Foods Int'l. Inc., and Cambria Quartz Surfaces. Designed to milk approximately 3,200 Jersey cows and house 1,000 Jersey dry cows, heifers, and baby calves, NSD is home to approximately 4,500 animals. The well-being of the animals is a key focus of dairy personnel because happy, healthy, comfortable cows are required for a dairy business to have success; which is defined, in part, as high-volume production of high-quality milk. Proper nutrition and feeding practices are two more essential requirements necessary for maintaining happy, healthy, comfortable cows.



NUTRITION

At NSD, cows are fed once per day, utilizing a Total Mix Ration (TMR), which incorporates all the necessary nutrients for cow health and productivity. It is essential that all of the proper nutrients are present in correct proportions to optimize the health of the cow and production of high-quality milk. These nutrients are provided through the use of a variety of agricultural commodities grown locally, regionally and even nationally based upon suitability, availability and economics. In addition, micronutrients including vitamins and trace minerals are purchased from animal nutrition companies. NSD utilizes the corn, corn silage, soybean meal, hay, and straw from over 10,000 acres per year. This allows the dairy to add value to these commodities locally while simultaneously lowering the distance, cost of delivery, environmental impact of transportation and minimize any negative aspects of transportation on infrastructure in the process of getting the crops to market. No drugs or hormones are fed to the cows at

NSD and each cow eats approximately 80 lbs. of feed per day. Water is provided in drinking fountains throughout the facility to access whenever the cows desire. Several technologies are employed to provide high quality, conditioned water in convenient locations and unlimited quantities for the benefit of the cows. Each cow will consume between 27 and 35 gallons of water each day depending on the weather.



MANURE MANAGEMENT

Any living species that eats produces waste and in cows this is commonly called manure. Here at NSD dairy, it is not thought of as manure or waste, but as a raw material providing nutrients to the farmers who grow the crops the dairy purchases for feed.

The cows at NSD generate approximately 100,000 gallons of “nutrients” each day. These nutrients are vacuumed up into a tank 3 times a day, each time a pen of cows goes to the parlor to get milked. This provides the cow with a clean environment for her health and comfort. The nutrients are then pumped into a flume where recycled water carries them to a sand separation lane. The sand bedding that is used in the barns for bedding the cows is separated from the nutrients and reclaimed for reuse from the sand separation lane. This practice keeps the barns as clean as possible and preserves the nutrient value in the liquid. The nutrients flow to one of 2 solids separation basins where the fiber-based solids settle, leaving mostly water to fill the larger covered storage basins. It is this “gray water” that is re-circulated back to the head of the flume where it carries nutrients back to the sand lane and basins. This process is ongoing and is utilized to continuously transport and store the nutrients. In the fall of the year after the crops are harvested, the basins are mixed together and the resulting nutrient-rich liquid is injected into the ground where it will

feed the crops planted in the spring. These nutrients are injected in quantities that supply exactly what next season’s crops need in the way of nitrogen, potash and phosphorus which saves the grower money, reduces the use of chemical fertilizer and prevents surplus nutrients from building up in the soil and compromising surface and ground water. Approximately 2,400 acres around NSD receive these nutrients each year. Injecting these nutrients in a sensible, agronomically responsible manner is good for the dairy, the grower, and the environment!

MILKING PARLOR

NSD has a 72 stall rotary milking parlor. Every 7 seconds a cow steps on to the rotating deck into an individual stall. Each cow is identified by a Radio Frequency Identification (RFID) tag in her ear. The computer identifies which cow is in which stall, when she comes in, when she leaves, when she starts and stops milking, and how much milk she gave. This information is utilized by management to monitor the cow preparation procedures and as indications of the cow’s health. Antiseptic spray is applied to each cow’s teats to prevent infection, the teats are wiped clean and dry, the milking units are attached and begin to perform the sucking action that stimulates the milk to come out of the cow’s udder. The milk harvest takes about 5 to 6 minutes. This takes place 3 times a day. Jersey cows give about 7 gallons of milk per day with a fat content of approximately 4.8% and a protein of 3.5%. In contrast, whole milk from the store contains 3.3% fat and 3.0% protein. The milk is then filtered and cooled to 37 degrees Fahrenheit and pumped directly onto a milk tanker for transport to the cheese factory. This process helps ensure that the milk is wholesome and fresh just like the products that will be manufactured from it.

The milking parlor is thoroughly cleaned every 8 hours. Cleaning is done to keep the equipment sanitary and the environment in the parlor clean in order to produce the highest quality milk.

SPECIAL NEEDS BARN

During the time when a cow is preparing to give birth to her calf she has special needs. The dairy has built special accommodations to best serve the animals during this time in their life. The cows are observed while giving birth and are assisted, if it is deemed necessary. Most cows give birth without assistance from staff. Cows that have recently calved will be housed here and closely monitored. Most cows give birth approximately once per calendar year and, since NSD is where both Northern Plains Dairy and NSD’s cows give birth, there are about 6,000 calves born at NSD each year.